

Day 20 Notes – Introduction to Terraform & Environment Setup

1. What is Terraform?

Terraform is an **Infrastructure as Code (IaC)** tool developed by **HashiCorp** that allows us to create, manage, and automate cloud infrastructure using code.

Benefits of Terraform

- Infrastructure as Code (IaC)
- Multi-cloud support (AWS, Azure, GCP)
- Automation of infrastructure provisioning
- Version control for infrastructure
- Consistent and repeatable deployments

Terraform Workflow

Write Code → terraform init → terraform plan → terraform apply → Resources Created

2. Tools Required

1. Terraform

Used for provisioning cloud infrastructure.

2. Git Bash

Provides Linux-like command-line experience on Windows.

3. AWS CLI

Used to interact with AWS services from the command line.

4. AWS Account

Required for creating cloud resources.

3. Install Git Bash

Download Git

Official Website:

<https://git-scm.com/downloads>

Installation Steps

1. Download Git for Windows
2. Run the installer
3. Click Next through default options
4. Finish installation

Verify Installation

`git --version`

Expected Output:

`git version 2.xx.x`

4. Install Terraform

Download Terraform

Official Website:

<https://developer.hashicorp.com/terraform/downloads>

Download:

Windows AMD64

Installation Steps

1. Download ZIP file
2. Extract ZIP
3. Create folder:

`C:\Terraform`

4. Copy:

`terraform.exe`

to:

C:\Terraform

Add Terraform to PATH

1. Search "Environment Variables"
2. Open Environment Variables
3. Select Path
4. Edit
5. Add:

C:\Terraform

6. Click OK

Verify Installation

terraform -version

Expected Output:

Terraform v1.x.x

5. Install AWS CLI

Download AWS CLI

Official Website:

<https://aws.amazon.com/cli/>

Installation Steps

1. Download AWS CLI v2
2. Run AWSCLIV2.msi
3. Click Next
4. Install
5. Finish

Verify Installation

aws --version

Expected Output:

aws-cli/2.x.x

6. Create IAM User in AWS

Why IAM User?

Instead of using the Root Account, Terraform should use an IAM user with appropriate permissions.

Steps

1. Login to AWS Console
2. Open IAM Service
3. Click Users
4. Click Create User

Example:

terraform-user

5. Click Next

Attach Permissions

For Lab Practice:

AdministratorAccess

6. Click Create User
-

7. Create Access Key & Secret Key

Steps

1. Open IAM User
2. Select Security Credentials Tab
3. Scroll to Access Keys
4. Click Create Access Key

Select Use Case

Command Line Interface (CLI)

5. Click Next

6. Click Create Access Key

AWS generates:

Access Key ID

Secret Access Key

Example:

AKIAxxxxxxxxxxxxxxxx

xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

Important

Download the CSV file or save the credentials securely.

8. Configure AWS CLI

Run:

aws configure

Provide:

AWS Access Key ID

AWS Secret Access Key

Default Region Name

Output Format

Example:

aws configure

AWS Access Key ID [None]: AKIAxxxxxxx

AWS Secret Access Key [None]: xxxxxxxxx

Default region name [None]: us-east-1

Default output format [None]: json

9. Verify AWS CLI Connectivity

Run:

```
aws sts get-caller-identity
```

Expected Output:

```
{
  "UserId": "AIDAxxxxxxx",
  "Account": "123456789012",
  "Arn": "arn:aws:iam::123456789012:user/terraform-user"
}
```

This confirms AWS CLI is successfully authenticated.

10. Verify Terraform Connectivity to AWS

Create a folder:

```
mkdir terraform-demo
```

```
cd terraform-demo
```

Create:

```
main.tf
```

Add:

```
provider "aws" {
  region = "us-east-1"
}
```

Initialize Terraform:

```
terraform init
```

Expected Output:

Terraform has been successfully initialized!

Key Commands Summary

Git

`git --version`

Terraform

`terraform -version`

`terraform init`

`terraform validate`

`terraform plan`

`terraform apply`

`terraform destroy`

AWS CLI

`aws --version`

`aws configure`

`aws sts get-caller-identity`

`aws s3 ls`

Interview Questions

Q1. What is Terraform?

Terraform is an Infrastructure as Code (IaC) tool used to provision and manage infrastructure through code.

Q2. What is IaC?

Infrastructure as Code is the process of managing infrastructure using configuration files instead of manual actions.

Q3. What is the purpose of terraform init?

It initializes the Terraform working directory and downloads required provider plugins.

Q4. Why do we use AWS CLI?

AWS CLI allows users to interact with AWS services from the command line.

Q5. Why should we use IAM users instead of the Root Account?

IAM users provide secure, controlled access and follow AWS security best practices.

Q6. What is the purpose of Access Key ID and Secret Access Key?

They are used by AWS CLI, SDKs, and Terraform to authenticate and access AWS services securely.